

Data Structure Assignment #1

1. Assignment

- 1) Implement dynamic arrays and singly linked lists with its functionalities.
- 2) Measure performance of each data structure and write a report on this.
- 3) Compress the report, source code, and executable file into a single ZIP archive.
- 4) E-class Submission: Upload the ZIP file to the designated e-class assignment.

2. Assignment details

This assignment delves into the world of data structures, specifically focusing on dynamic arrays and linked lists. Throughout this exploration, you'll gain a hands-on understanding of their functionalities and identify the strengths and weaknesses of each approach.

The core objective of this assignment is to conduct a thorough comparison between dynamic arrays and linked lists. Here's a roadmap for your investigation:

1) Requirements

- The implementation must be in C language.
- Source code must be compliable.
- The code must be clear and concise (proper indentation, meaningful variable names, and comments).

2) Assignment #1-1: Implementation

Implement a dynamic array and a singly linked list. Requirements are as follows:

- Implement a dynamic array data structure with functions for:
 - Initialization with a specific size
 - Inserting an element (front, back, specific index)
 - Deleting an element (front, back, specific index)
 - Accessing an element at a specific index
- Implement a singly linked list data structure with functions for:
 - Initialization (empty list)
 - Inserting an element (front, back, specific index)
 - Deleting an element (front, back, specific index)
 - Accessing the element at the head
- On the report, explain each implementation and examples.

3) Assignment #1-2: Time Complexity Analysis:

Observe the time complexity of each operation of both data structure. Requirements are as follows:

- Write code to measure the execution time of each operation (insertion, deletion, access) for both dynamic arrays and linked lists.
- Vary the number of elements in the data structure (e.g., 10, 100, 1000, 10000) and record the execution times for each operation at each size.
- On the report, analyze the results and determine the time complexity (Big O notation) of each operation for both data structures. Explain your reasoning.

4) Assignment #1-3: Space Complexity Analysis:

- Estimate the space complexity (in terms of memory usage) of each data structure. Consider the size of the data element itself and any additional overhead for storing metadata (e.g., array size, pointers in linked lists).
- Explain how the space complexity of each data structure relates to the number of elements stored (Hint: dynamic arrays multiple it's size).

5) Report

- Based on your findings, compare the time and space complexity of dynamic arrays and linked lists for insertion, deletion, and access operations.
- Discuss the scenarios where each data structure would be a better choice based on its time and space complexity characteristics.
- Discuss the trade-offs between these factors and how they influence the choice of data structure for specific programming tasks.